Laxy Studio Function List

Laxy Studio — Function List

> **Hierarchy:** Module → Sub-Module → Function → Feature → Ticket → Remark
> **Scope:** Laxy Studio (B2B creator platform). Player PWA = separate table.
> **References:** Local docs 01–06 + Confluence: System Architecture, Development Roadmap

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Proposed Complete Table

Module	Sub-Module	Function	Feature	Phase	Ticket	Remark
Infrastructure	Auth & RBAC	SSO / OAuth Login	• Google / Apple social login - Firebase Authentication	Phase 1		ok, just like what TMX did, so we also need use resend.com (need another account for free plan)
		JWT / Session Management	• Issue & refresh access tokens - Token expiry policy	Phase 1		will implement
		Role-Based Access Control	• Roles: Laxy Super Admin, Client Admin, Client Editor - Permission matrix per role	Phase 1		yes, can implement by Firebase Auth, but need you define the **permission matrix**

						HL: System Architecture | 4. User Roles & Permissions Use FireCMS to implement
	Cloud Storage	Asset Storage	• Support PDF, image, video, audio ~~- Tenant-scoped isolation~~	Phase 1		GCP Cloud Storage **Tenant scoped,** so it should be equivalent to “Client” in RBAC. You decide align the term or not HL: Client should also control asset ⇒ Tenant > Client Admin
		Output Storage	• Store generated audio, SRT, slideshow assets - Versioned output	Phase 2		How the versioning **pattern** be? just v1,2,3? HL: please design

						Need more detail > Phase 2
	Database	Relational DB (PostgreSQL)	• Users, ~~Tenants,~~ Permissions, Billing, Prompts	Phase 1		PostgreSQL, yes, also fit for flowise Again, • **Users, Client Admin terms are bit confusing?** HL: let’s discuss
		Document DB (Firestore)	• Content, Spots, Scripts, AudioFiles (real-time sync)	Phase 1		Just Firestore.
		Vector DB	• AI embeddings for Chatbot & Q&A semantic search	Phase 3		**Should not be in Phase 1** Pinecone / Weaviate — Phase 3 dependency HL: yes phase 3
	API Gateway	API Routing & Auth Middleware	• Control by flowise	Phase 1		Most control by flowise on Phase 1, can set gateway

						in next phases All AI ops return taskId for async polling > @Sun draw a clear relationship between API
			• Single entry point, Firebase ID token validation - Base URL: `api.laxy.travel/v1`	Phase 2		
		Rate Limiting & Versioning	• Per-tenant rate limits - Versioned API contracts	Phase 1		**confirm set in Phase 1?** it is a question, since for Gemini and 3rd party side, also has rate limit. so need some testing first

						• Per-tenant rate limits, the first assumption is we can have much huge rate limit in Gemini side first Display token spending for internal evaluation first
	AI Orchestrator	Event Queue (Pub/Sub)	• Async routing between Studio and Orchestrator - Key events: ASSET_UPLOADED → CONTENT_PUBLISHED	Phase 1		Flowise queue mode
		Prompt Router	• Route prompts by function via Prompt Library - Vendor-agnostic model adapter	Phase 1		I think the prompt node in flowise is static in Phase 1 which set to hard coded to specific LLM Decoupled — swap AI

						providers without code change HL: Hard coded to Gemini 3.1 pro preview? Or could define on which model by on each prompt in library?
		NLP / OCR Service	• Text/PDF/Excel/Word only	Phase 1		yes, and faced rate limit before, need apply for larger limit HL: Cost eatimation. Have to implement in different phase?
			• Images	Phase 2		
		Script & Translation Service	• Gemini Pro: generate and translate scripts	Phase 1		yes
		TTS Service	• Gemini 2.5 Pro TTS: generate speech audio	Phase 1		yes
		Image Generation Service	• Gemini / Imagen: avatars,	Phase 2		yes HL: Do in phase 2

			slideshows, stamp art			
	CI/CD & Environments	Deployment Pipeline	• GitHub Actions: Studio, Player, API, Orchestrator pipelines - Environments: Dev / Staging / Production	Phase 1		laxy-pipelines (Orchestrator pipelines) laxy-studo (included FireCMS(laxy-studo/admin)) laxy-guide (React Frontend Player) **so please confirm the URL:** [env]-studio.laxy.travel [env]-guide.laxy.travel/{guideID} **confirm the flow:** DEV, Staging, PROD

						data are totally isolated No data migration at normal case. **Expect tenant only do operation in PROD studio?** Different env TOTALLY separate
		Monitoring & Logging	• Cloud Logging, Monitoring, Error Reporting - Audit logs for all roles actions	Phase 1		Phase 1, by default **show what flowise default provide?** Internal Laxy team HL: Audit logs for all roles actions → All users
	Security	Data Protection	• Encryption at rest and in transit (TLS) - • ~~Signed URLs for file access~~	Phase 2		• Prompts fetched server-side only, yes, only in

			• Prompts fetched server-side only • ~~GCP Secret Manager for secrets~~			flowise, not in studio No handle signed URLs in Phase 1 No special GCP Secret Manager, just follow Firestore practice in Phase 1
			• Signed URLs for file access • GCP Secret Manager for secrets	Phase 2		
Super Admin	Prompt Library	Prompt Management	• Create, version, activate AI prompts per function • Prompt tagging by module • Only ~~Client~~ Super Admin can edit selected prompts	Phase 1		yes, Decouple AI tuning from code deployment it should all the flowise can provide HL: Only Super Admin could edit prompt
		A/B Prompt Testing	• Run prompt variants, compare output quality	Phase 2		Option 1: Same Pipeline + Different Variable/Override Values

						Option 2: Different Pipelines (Recommended for Bigger Variants) Multi prompt under same module for selection
	User & Tenant Management	Tenant CRUD	• Create, read, update, deactivate tenant accounts	Phase 1		Recommended high-level architecture for Laxy Studio Phase 1 MVP laxy-studio (React frontend + FireCMS) - Main app: studio.laxy.travel - **Admin: /admin** route with **FireCMS** - Wizard for Guide creation
		User CRUD	• Manage users within tenants • Assign roles (Client Admin / Client Editor)	Phase 1		Handle in FireCMS

		Subscription Tier Override	• Manually adjust tier/quota for a tenant	Phase 1.5		Handle in FireCMS
		Module Dashboard	• Enable/disable modules per tenant - Controls Studio and Player visibility	Phase 2		Handle in FireCMS
	Platform Config & Analytics	Feature Flags	• Enable/disable features per tenant or globally	Phase 1		Handle in FireCMS
		Platform Analytics	• Usage metrics and revenue overview across all tenants	Phase 3		Handle in FireCMS
Laxy Studio	Remark		AI would make mistake. Please verify before publish			
	Member Management	Login	• Social login (Google, Apple) - Email & password login - Email OTP verification	Phase 1		Handle in Studio App
		Registration	• Tenant sign-up form - Email verification - Onboarding walkthrough	Phase 1		Handle in Studio App

		Forget Password	• Reset via email link - OTP expiry policy	Phase 1		Handle in Studio App
		Account Management	• Change email, password, display name - Profile avatar upload	Phase 1		Handle in FireCMS
		Tenant Information	• Company name, logo, billing info - Timezone & default language - Default audio settings (Character, Voice, Context) used by Quick Process	Phase 1		Handle in FireCMS
		User Management	• Invite users by email - Assign / change roles - Deactivate / remove users	Phase 2		Handle in FireCMS
	Asset Management	Asset Upload	• Drag & drop upload (PDF, image) • Bulk upload with progress indicator • URL / raw text input as content source	Phase 1		Handle in Studio App (Wizard) Max file size TBD

						HL: 100M
			• Drag (video, audio)	Phase 2		
		Asset Library	• Browse, search, preview assets • Delete ~~/ replace~~ assets • Storage usage indicator	Phase 1		Handle in Studio App (Wizard) HL: no relpace
		~~Asset Tagging~~	• ~~Tag assets for retrieval across modules~~			Not in Phase 1? HL: no tag need, I thnk
	Basic Information	Entity Configuration	• Venue name, address, GPS coordinates • Map image, cover image • Operating hours, website, phone • Supported languages • Core language • Enabled modules • Item field config (e.g. title, year...)	Phase 1		Handle in Studio App (Wizard) Top-level setup before any module
		Layout Style Selection	• Select from pre-designed Player UI templates (Phase 1)	Phase 1		Handle in Studio App (Wizard)

			• Live preview of selected layout			
		AI Layout Generation	• Generate custom UI from brand reference + prompt	Phase 3		Phase 2
		Function / Module Selection	• Guide	Phase 1		Managed via Module Dashboard
			• Q&A • Chatbot • Stamp Hunt	Phase 3		
	Guide — Step 1: Ingestion	Content Upload	• Select files already in Asset Data Lake, OR upload new files here (New uploads automatically saved to Asset Data Lake • ~~Or provide basic entity info manually~~ • ~~Select core language~~	Phase 1		Handle in Studio App (Wizard) and Flowise Files always centralised in Asset Data Lake regardless of entry point

		AI Metadata Extraction	• AI extracts: Title, Artist, Period, Material, Dimensions, Highlight, Cultural Designation, and more • Add number to items (if any) • Drag-and-drop reorder	Phase 1		Handle in Studio App (Wizard) and Flowise Gemini Multimodal OCR/NLP
		Human Gate 1 — Data Review	• Client reviews and corrects extracted metadata inline • Approve to proceed	Phase 1		Handle in Studio App (Wizard) and Flowise
			• Export to Excel for batch review	Phase 2		
	Guide — Step 2: Deep Research *(Phase 2)*	Deep Research Activation	• Optional: only if Deep Research module is enabled • **Degree of Research selection**: Low / Medium / High (higher degree = broader crawl but higher risk of inaccurate content) • AI crawls: art critiques,	Phase 2		Phase 2 — Epic 7

			historical backgrounds, artist biographies, conservation notes - ⚠️ Client must be clearly informed: higher depth increases inaccuracy risk mandatory Human Gate 2 review before proceeding			
		Human Gate 2 — Research Review	• Client reviews AI-researched content for accuracy - Approve to proceed	Phase 2		Skipped in Quick Process
	Guide — Step 3: Script Generation	AI Script Generation	• AI generates polished rewritten description per spot • Based on approved metadata	Phase 1		Handle in Studio App (Wizard) and Flowise Multiple variants = different end-user depth options
			• AI generates audience-variant scripts per spot:	Phase 2		

			◦ Kids, Academic, Quick (1-min), Professional, Brief ◦ In-Depth / Enthusiast (requires Deep Research, Phase 2) ◦ V1, V2, V3 variants per type			
		Image-Spot Mapping	• Assign images/videos from asset library to each spot ~~- AI auto-suggests media~~ ~~- Manual override~~	Phase 1		??? how the logic would be??? **I guess it is hard to get good result** HL: AI assign too complicated
		Human Gate 3 — Script Review	• Client reviews AI-generated scripts per spot - Approve individually or in bulk • **Fast Track**: bypass gate for trusted/simple items	Phase 1		Fast Track = spot-level skip; Quick Process = full pipeline bypass Handle in Studio App (Wizard) and Flowise

	Guide — Step 4: Translation	AI Translation	• AI translates approved base script to all configured languages (EN, JP, KR, ZH-TW, ZH-CN, FR, etc.) • Culturally appropriate phrasing	Phase 1		Handle in Studio App (Wizard) and Flowise Gemini Pro
		Human Gate 4 — Translation Review	• Client reviews translated scripts • Approve to proceed	Phase 1		Handle in Studio App (Wizard) and Flowise Skipped in Quick Process
			• Export to Excel for external translators	Phase 2		
	Guide — Step 5: Audio	Character Selection	• Select from Character Library (voice persona: name, role, avatar, personality, speech patterns) • ~~Client Admin: create new character via meta prompt~~	Phase 1		Handle in Studio App (Wizard) and Flowise Check this Youtube link before these magic happen 那些 Agent 神文没告诉你的事：照着做，系统只会更烂 【AI ag

						ent 搭建实操指南】 HL: ⇒ Sure not everything is AI, instead most of ity is not AI Character ≠ TTS voice — it is a full voice recipe
			• ~~AI recommends best-fit character (⭐ Suggested), according to entity Configuration~~	~~Phase 3~~		~~POC needed~~
		Voice Selection	• Filter by: All / Female / Male • AI recommends based on entity configuration (non-AI, only by logic) • Available: Aoede, Algieba, Algenib, Despina, Laomedeia, Pulcherrima, Sadaltager, Sulafat	Phase 1		Handle in Studio App (Wizard) and Flowise Gemini 2.5 Pro TTS voices

		Context / Director Note	• Optional situational directive • AI generates: Vocal Environment, Mission of Speech, Pacing & Energy • Admin can edit underlying meta-prompt	Phase 1		Handle in Studio App (Wizard) and Flowise
		Script Enhanceme nt	• Toggle ON/OFF (default OFF) • AI adds performance cues: non-speech sounds, style modifiers, pacing tags • Max 2 tags per sentence • Client edits enhanced script per language tab and can regenerate	Phase 2		Phase 2 — Epic 8.1
		Audio Generation	• Generate audio per language or all languages • Token estimate shown before generation - Progress bar	Phase 1		Handle in Studio App (Wizard) and Flowise Japanese Kanji→Hiraga

			during generation • **Japanese**: Kanji → Hiragana conversion before TTS • **Chinese (TW/CN)**: pronunciation substitution library			na critical for audio quality
		Audio Playback & QA	• Audio player per language tab: play, pause, skip ±15s • Admin: download .mp3 (24000Hz) • Production: audio written directly to CMS	Phase 1		Handle in Studio App (Wizard) and Flowise
		Human Gate 5 — Audio Review	• Client listens per language	Phase 1		Handle in Studio App (Wizard) and Flowise
			• Edit enhanced script and regenerate if did script enhancement	Phase 2		
		Audio Pronunciation Fix *(POC Needed)*	• Client marks a specific timestamp + leaves a text comment	Phase 1		**Smart Re-generation in Phase 2**

			describing the voice issue • Remark that **"This is AI function and can make mistakes. Please check carefully before launch.**			Feasibility to be confirmed — surgical re-gen requires precise audio stitching
			• **AI re-generates only that audio segment and blends it back into the full audio** • Reference POC built in Google AI Studio: [link]	Phase 2		HL: Have to try more. So far Google AI Studio version not 100% work
		Generation History	• Every run saved: timestamp, languages, character, context, token count • Click to retrieve scripts and audio for auditing, rollback, and cost tracking	Phase 1		Handle in Studio App (Wizard) and Flowise
		SRT Generation	• Auto-generate subtitle files	Phase 1		

			• synced to audio timing • With the original script as base			
	Guide — Step 5: Publishing	Slideshow Builder (TTML)	• **Phase 1 (Manual):** Client selects images/videos from Asset Management per spot • Multi-image selection supported per spot - Guide overview: visual timeline editor and adjust image order and sync timing	Phase 1		**From This part, v2 now in Decap CMS. so what is the plan to merge it to Laxy Studio?** Phase 1 = manual selection
			• AI auto-generates and sequences images synced to audio per spot (AI Layout Generation)	Phase 2		Phase 2 = AI layout generation — Epic 8.4
		CMS Publishing	• Final content check before publish • Bundle approved content (audio, scripts, images, SRT)	Phase 1		Final pre-publish verification step

			• Push to Firebase CDN			
		Preview	• Client previews final Player experience • Select / confirm Player template	Phase 1		
		Final Approval & Publish	• A preview in mobile view shows • Final review all page and content	Phase 1		
	Guide — Quick Process *(Phase 2)*	Quick Process Mode	• One-click batch using Tenant default settings (Character, Voice, Context) • Skips Human Gates 2, 3, 4 • Auto-generates all language variants - Pushes to "Pending Review" for final Gate 5 only	Phase 2		Phase 2 — Epic 8.5. Defaults configured in Tenant Information
	Publishing & Delivery	QR Code Generation	• Auto-generate QR code on publish • Downloadable as PNG	Phase 1		Short URL critical for clean QR
		Shortlink / URL	• Generate branded short	Phase 1		Critical dependency

			URL (laxy.click) • Custom slug option			for QR quality
		Embed Code	• Generate embed snippet for third-party websites	Phase 3		
		Unpublish / Archive	• Deactivate live experience • Archive with data retained	Phase 2		
	Analytics & Insights	Experience Analytics	• Views, unique visitors, completion rate • Per-spot engagement: time spent, drop-off, audio play rate • Script variant popularity	Phase 1		Phase 2 — POC required before committing to build Take a look on cost
			• **GA Integration, POC Needed):** Connect with Google Analytics (each Player page has a GA seed) to pull and display GA data inside Studio dashboard • AI-assisted analysis layer: summarise	Phase 2		

			insights and surface recommendations • ⚠️ Cost investigation needed: analyse data in Laxy Studio vs. leverage Google Analytics native tools and select more cost-efficient path			
		Export Report	• Download as CSV or PDF	Phase 1		
	Billing & Subscription	Subscription Plan	• Manual plan setup • View current plan and limits (experiences, storage, users) • Feature availability by tier	Phase 1		**handle in FireCMS** **no idea the scope of phase 1**
			• Integrate Payment gateway	Phase 3		
		Plan Upgrade / Downgrade	• Self-serve plan change with prorated billing	Phase 3		
		Usage Meter	• Real-time quota usage display	Phase 2		

		Invoice & Payment History	• View and download invoices	Phase 2		
	Notification	Email Notification	• New user invite, publish success, billing alerts	Phase 2		

Phase Summary

Phase	Module / Feature
Phase 1 (MVP)	Infrastructure, Super Admin, Member Management, Asset Management, Entity Config, Guide Steps 1–5 (core, manual), Slideshow Builder (manual), Publishing & Delivery
Phase 2	Deep Research, Script Enhancement, SRT, Slideshow Builder (AI-Assisted / AI Layout Generation), Quick Process, GA Analytics Integration
Phase 3	Chatbot, Q&A, Stamp Hunt
POC Needed	Audio Pronunciation Fix (segment re-gen + stitch), GA Analytics AI Analysis Layer

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Open Items

1. **Player PWA** — Separate function list table.

2. **Voice list** — Finalise available voices per Kazu's research.
3. **Voice Cloning** — Future feature. Licensing required — Alex to confirm with sales team.
4. **Japanese Kanji → Hiragana** — Critical pre-processing step for audio quality. Done transparently server-side.
5. **Chinese Pronunciation Library** — Maintainable substitution table for ZH-TW / ZH-CN edge cases.
6. **Shortlink / URL Service** — Critical for clean QR codes. Third-party service to be selected.
7. **Audio Pronunciation Fix (POC)** — Segment-level audio re-generation and stitching. Feasibility TBD. Reference POC built in Google AI Studio by Harold.
8. **Deep Research Degree UX** — Define exact UI for Low / Medium / High selection. Warning copy to be reviewed by Alex.
9. **GA Analytics Integration (POC)** — Investigate cost: build in-Studio analytics pipeline vs. surface GA data directly. Decision to be made before Phase 2 build.